

A sharp dichotomy at $\ell \log \ell \asymp \log Q$ for joint-family log-free zero-density in cyclotomic Kummer Hecke towers

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Abstract

Let $\chi_{\ell,A}$ be the rank-two cyclotomic Kummer Hecke character on $K_\ell = \mathbb{Q}(\zeta_\ell)$ attached to data $(\ell, A) \in \{\text{prime}\} \times \mathbb{F}_\ell^2 \setminus 0$, cutting out the Kummer extension $K_{\ell,A} = K_\ell(2^{A_1}3^{A_2})^{1/\ell}$. We prove an unconditional sharp dichotomy for the joint-family log-free zero-density estimate at threshold $\ell \log \ell \asymp \log Q$.

Unconditional unlock for $\ell \leq \log Q / \log \log Q$. Setting $\eta \asymp 1/\log Q$ and $T \leq (\log Q)^{B'}$, the joint family sum

$$\sum_{\ell \leq \log Q / \log \log Q} \sum_{A \in \mathbb{F}_\ell^2 \setminus 0} N_{\chi_{\ell,A}}(1 - \eta, T) \ll (\log Q)^C$$

holds unconditionally, by trivial concatenation of the Thorner–Zaman 2019 log-free zero-density estimate per character. In particular the high-seam Prime-Hecke-Non-Concentration (PHNC) bound

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, \chi_{r,s}(q) = 1\} \leq (1 - \ell^{-C(B)})\pi(Q; \ell, 1)$$

holds uniformly for $\ell \leq \log Q / \log \log Q$, $(r, s) \neq 0$, unconditionally.

Unconditional no-go for $\ell \geq \log Q$. Conversely, for any $B > 1$, the joint-averaged log-free ZDE for $\ell \leq (\log Q)^B$ fails to follow from any combination of Pierce–Turnage–Butterbaugh–Wood 2020 (Duke), Weiss 1980/1983 (Crelle), Thorner 2017 (IMRN), Thorner–Zaman 2019 (Alg. Number Theory), or Heath-Brown 1992 (PLMS). The obstruction is a structural Cross-Field Rankin–Selberg Conductor Explosion: any large-sieve majorant inherits an off-diagonal cost $\geq \exp(\Omega(X^2 \log X))$ from the Bushnell–Henniart pair conductor $\mathfrak{q}(\chi_{\ell_1, A_1} \otimes \overline{\chi_{\ell_2, A_2}}) \leq (\ell_1 \ell_2)^{\ell_1 \ell_2}$, super-polynomial in the family size $|\mathcal{F}(X)| \asymp X^3 / \log X$. The Artin descent $\text{Ind}_{K_\ell}^{\mathbb{Q}} \chi_{\ell,A}$ provides no structural rescue.

Counterexample refutation. No Landau–Siegel real zero of $L(s, \chi_{\ell,A})$ can drive a counterexample for $\ell \leq V_{\text{Vandiver}} = 1.63 \times 10^8$ (Buhler–Harvey 2011), because the Kummer character $\chi_{\ell,A}$ has odd prime order $\ell \geq 3$, hence is non-real; the only real component of the Aramata–Brauer factorization of $\zeta_{K_{\ell,A}}$ is the Legendre symbol mod ℓ , disjoint from the Kummer family.

1 Setup and notation

Fix a prime $\ell \geq 3$ and a pair $(a_1, a_2) = (2, 3)$. For $A = (A_1, A_2) \in \mathbb{F}_\ell^2 \setminus 0$, write

$$\alpha_A = 2^{A_1} 3^{A_2} \in \mathbb{Q}^\times, \quad K_\ell = \mathbb{Q}(\zeta_\ell), \quad K_{\ell,A} = K_\ell(\alpha_A^{1/\ell}).$$

Kummer theory and rational Kummer independence of $\{2, 3\}$ over $\mathbb{Q}(\zeta_\ell)$ (cf. companion paper *Rank- r synchronization defects and projective Kummer geometry*) give $[K_{\ell,A} : K_\ell] = \ell$, with $\text{Gal}(K_{\ell,A}/K_\ell) \simeq \mathbb{Z}/\ell$. The Kummer character $\chi_{\ell,A}$ is the non-trivial ℓ -th-power Hecke character of K_ℓ cutting out $K_{\ell,A}$.

The absolute discriminant of $K_\ell/\mathbb{Q}$ is $d_{K_\ell} = \pm \ell^{\ell-2}$ (Apostol/Washington Prop. 2.7), so

$$\log |d_{K_\ell}| = (\ell - 2) \log \ell \asymp \ell \log \ell.$$

The conductor-discriminant formula gives the Artin conductor of the induced representation

$$\text{cond}(\text{Ind}_{K_\ell}^{\mathbb{Q}} \chi_{\ell,A}) = |d_{K_\ell}|^{\dim \chi_{\ell,A}} \cdot N_{K_\ell/\mathbb{Q}}(\mathfrak{f}(\chi_{\ell,A})) = \ell^{\ell-2} \cdot N(\mathfrak{f}),$$

with $N(\mathfrak{f}(\chi_{\ell,A})) \leq (\ell \cdot N(\alpha_A))^{\ell-1}$, bounded by an absolute constant for our fixed $(a_1, a_2) = (2, 3)$.

The analytic conductor in the Iwaniec–Kowalski normalization is

$$\mathfrak{q}(\chi_{\ell,A}, T) = |d_{K_\ell}| \cdot N_{K_\ell/\mathbb{Q}}(\mathfrak{f}(\chi_{\ell,A})) \cdot (3 + |T|)^{\ell-1}.$$

Let $N_\chi(\sigma, T) = \#\{\rho : L(\rho, \chi) = 0, \Re \rho \geq \sigma, |\Im \rho| \leq T\}$ counted with multiplicity. The joint family is

$$\mathcal{F}(X) = \{\chi_{\ell,A} : \ell \leq X \text{ prime}, A \in \mathbb{F}_\ell^2 \setminus \{0\}\}, \quad |\mathcal{F}(X)| = \sum_{\ell \leq X} (\ell^2 - 1) \sim \frac{X^3}{3 \log X}.$$

2 The unconditional log-free ZDE input

The starting point is the Thorner–Zaman log-free zero-density estimate.

Theorem 1 (Thorner–Zaman 2019 / 2021, log-free ZDE). *There exist absolute constants $c_1, c_2, c_3, c_4 > 0$ such that for every Hecke character χ of a number field K with $n_K = [K : \mathbb{Q}]$, modulus $\mathfrak{f}_\chi$, and analytic conductor $\mathfrak{q}(\chi, T)$,*

$$N_\chi(\sigma, T) \leq c_1 \cdot (e^{c_2 n_K} \cdot |d_K| \cdot N_{K/\mathbb{Q}}(\mathfrak{f}_\chi) \cdot (3 + |T|)^{n_K})^{c_3(1-\sigma)}$$

holds uniformly for $1/2 \leq \sigma \leq 1$ and $|T| \geq 1$, with $\sigma \geq 1 - c_4 / \log(|d_K| N(\mathfrak{f}) T^{n_K})$. Equivalently,

$$N_\chi(\sigma, T) \ll \mathfrak{q}(\chi, T)^{c_3(1-\sigma)},$$

the implied constant absolute.

The proof, in Weiss 1983, Thorner 2017, and Thorner–Zaman 2019/2021, proceeds via a power-sum / Turán step, a mean value over Hecke characters, a Halász–Montgomery argument, and the Deuring–Heilbronn phenomenon for the exceptional zero — each step using constants independent of K, χ, T . The base-field dependence enters *only* through the analytic-conductor packaging $\mathfrak{q}(\chi, T)$, in which $|d_K|$ appears multiplicatively in the parenthesis raised to $c_3(1 - \sigma)$.

Substituting our family parameters $K = K_\ell$, $|d_K| = \ell^{\ell-2}$:

$$N_{\chi_{\ell,A}}(\sigma, T) \ll \ell^{(\ell-2)c_3\eta} \cdot N(\mathfrak{f})^{c_3\eta} \cdot (3 + |T|)^{(\ell-1)c_3\eta}, \quad \sigma \geq 1 - \eta.$$

The first factor is $\exp((\ell - 2)c_3\eta \log \ell)$, governed by the product $\ell \log \ell \cdot \eta$.

3 The unconditional unlock for $\ell \leq \log Q / \log \log Q$

Theorem 2 (Unconditional PHNC at the strict polylog threshold). *Let $Q \geq 3$, $B' > 0$, $\eta = c / \log Q$ for an absolute $c > 0$. For every prime ℓ with $\ell \log \ell \leq \log Q$ (equivalently $\ell \leq \log Q / \log \log Q$ up to absolute constants),*

$$\sum_{A \in \mathbb{F}_\ell^2 \setminus 0} N_{\chi_{\ell,A}}(1 - \eta, (\log Q)^{B'}) \ll (\log Q)^C \cdot \ell^2.$$

Summing over $\ell \leq \log Q / \log \log Q$,

$$\sum_{\substack{\ell \leq \log Q / \log \log Q \\ \ell \text{ prime}}} \sum_{A \in \mathbb{F}_\ell^2 \setminus 0} N_{\chi_{\ell,A}}(1 - \eta, (\log Q)^{B'}) \ll (\log Q)^{C'},$$

both inequalities unconditional, with C, C' absolute constants depending only on c and B' .

In particular the high-seam PHNC instance

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, \chi_{r,s}(q) = 1\} \leq (1 - \ell^{-C(B)})\pi(Q; \ell, 1)$$

holds uniformly for $\ell \leq \log Q / \log \log Q$, $(r, s) \neq 0$, unconditionally.

Proof. Apply [Theorem 1](#) to $\chi_{\ell,A}$ at $\sigma = 1 - \eta$:

$$N_{\chi_{\ell,A}}(1 - \eta, T) \ll \ell^{(\ell-2)c_3\eta} \cdot N(\mathfrak{f})^{c_3\eta} \cdot T^{(\ell-1)c_3\eta}.$$

For $\eta = c / \log Q$ and $\ell \log \ell \leq \log Q$,

$$(\ell - 2)c_3\eta \log \ell \leq c_3c \cdot \frac{\ell \log \ell}{\log Q} \leq c_3c,$$

so $\ell^{(\ell-2)c_3\eta} \leq e^{c_3c}$, an absolute constant. Similarly $T^{(\ell-1)c_3\eta} \leq (\log Q)^{B'(\ell-1)c_3\eta} \leq (\log Q)^{B'c_3}$ in the regime $\ell \leq \log Q / \log \log Q$, $T \leq (\log Q)^{B'}$. $N(\mathfrak{f})^{c_3\eta}$ is absorbed into a constant because $N(\mathfrak{f}) \ll \ell^{\ell-1}$, contributing the same $\ell \log \ell \eta \leq c$ exponent.

Summing over $A \in \mathbb{F}_\ell^2 \setminus 0$ gives a factor $\ell^2 - 1$. Summing over $\ell \leq \log Q / \log \log Q$ gives a factor $(\log Q / \log \log Q)^{O(1)} \ll (\log Q)^{O(1)}$. The PHNC instance follows by the standard zero-density-to-Prime-Hecke-non-concentration conversion (Thorner–Zaman 2019 §5; Iwaniec–Kowalski Ch. 5.7). $\square$

Remark 1. [Theorem 2](#) is sharp at $\ell \log \ell = \log Q$: above this threshold the per-character bound contributes $\exp(\Omega(\ell \eta \log \ell))$ which blows up super-polylog. This is the unconditional content of the V16 BRIDGE corollary upgraded by V17 WTZ-Kum verdict.

4 The no-go theorem for $\ell \geq \log Q$

Theorem 3 (Joint-Averaging No-Go, JANG). *For every $B > 1$, the joint-averaged log-free ZDE*

$$\sum_{\ell \leq (\log Q)^B} \sum_{A \neq 0} N_{\chi_{\ell,A}}(1 - \eta, T) \ll_\varepsilon (\log Q)^{C\eta + \varepsilon} T^{c'\eta + \varepsilon}$$

uniform in Q, T does not follow from any combination of Pierce–Turnage–Butterbaugh–Wood 2020 (Duke), Weiss 1980/1983 (Crelle), Thorner 2017 (IMRN), Thorner–Zaman 2019/2021 (Alg. Number Theory), or Heath-Brown 1992 (PLMS).

The proof has three independent ingredients: [Theorem 4](#) (cross-field Rankin–Selberg conductor explosion), [Theorem 5](#) (per-character conductor explosion summed over ℓ), and [Theorem 6](#) (Artin descent does not help).

Theorem 4 (Cross-Field Rankin–Selberg Conductor Explosion). *Let $\mathcal{F}(X)$ be the cyclotomic Kummer joint family. Any large-sieve majorant of the form*

$$\sum_{\chi \in \mathcal{F}(X)} \left| \sum_{n \leq N} a_n \chi(n) \right|^2 \ll (C + N) \|a\|_2^2 (NC)^\varepsilon$$

obtained by Pierce–Turnage–Butterbaugh–Wood-style duality together with Rankin–Selberg off-diagonal control admits a uniformly nonzero off-diagonal contribution from cross- (ℓ_1, ℓ_2) pairs with

$$\mathfrak{q}(\chi_{\ell_1, A_1} \otimes \overline{\chi_{\ell_2, A_2}}) \leq (\ell_1 \ell_2)^{\ell_1 \ell_2 + O(1)}$$

by the Bushnell–Henniart conductor-of-pairs estimate. Consequently the off-diagonal cost is $\geq \exp(c \cdot X^2 \log X)$, super-polynomial in the family size $|\mathcal{F}(X)| \asymp X^3 / \log X$.

Proof. The PTW majorant for a family $\mathcal{G}$ of Hecke characters expands, after Cauchy–Schwarz and Mellin inversion, into a diagonal sum and an off-diagonal sum over pairs. The off-diagonal terms contain the Rankin–Selberg L -function $L(s, \chi_1 \otimes \overline{\chi_2})$ for distinct $\chi_i \in \mathcal{G}$, evaluated on a critical-strip line.

For $\chi_i = \chi_{\ell_i, A_i}$ on base fields $K_{\ell_i} = \mathbb{Q}(\zeta_{\ell_i})$, the tensor $\chi_1 \otimes \overline{\chi_2}$ lives on the compositum $K_{\ell_1} \cdot K_{\ell_2} = \mathbb{Q}(\zeta_{\ell_1 \ell_2})$ of degree $(\ell_1 - 1)(\ell_2 - 1)$ over $\mathbb{Q}$, viewed as a non-abelian Galois representation of dimension $(\ell_1 - 1)(\ell_2 - 1)$ over $\mathbb{Q}$ after induction.

By the Bushnell–Henniart 1997 (IMRN) local pair-conductor bound, applied to the wild-ramified factors at primes ℓ_1, ℓ_2 ,

$$a(\rho_1 \otimes \rho_2) \leq n_2 \cdot a(\rho_1) + n_1 \cdot a(\rho_2)$$

at every local place. Globally

$$\log \mathfrak{q}(\chi_{\ell_1, A_1} \otimes \overline{\chi_{\ell_2, A_2}}) \leq (\ell_2 - 1) \log \mathfrak{q}(\chi_{\ell_1, A_1}) + (\ell_1 - 1) \log \mathfrak{q}(\chi_{\ell_2, A_2}).$$

Substituting $\mathfrak{q}(\chi_{\ell_i, A_i}) \leq \ell_i^{\ell_i - 2} \cdot O(1)$:

$$\log \mathfrak{q}(\chi_1 \otimes \overline{\chi_2}) \leq (\ell_2 - 1)(\ell_1 - 2) \log \ell_1 + (\ell_1 - 1)(\ell_2 - 2) \log \ell_2 \asymp \ell_1 \ell_2 \log(\ell_1 \ell_2).$$

By the convexity bound for L -functions, $|L(1/2 + it, \chi_1 \otimes \overline{\chi_2})| \ll \mathfrak{q}^{1/4 + o(1)}$, so the off-diagonal contribution per pair is $\geq (\ell_1 \ell_2)^{\ell_1 \ell_2 / 4 - o(1)}$. Summing over pairs $\ell_1, \ell_2 \leq X$ gives the displayed bound. $\square$

Theorem 5 (Per-character conductor-explosion barrier). *Substituting $\mathfrak{q}(\chi_{\ell, A}) = \ell^{\ell - 2 + O(1)}$ into the Heath-Brown 1992 (PLMS) or Thorner–Zaman 2019/2021 per-character log-free ZDE and summing over the family yields an exponent $\geq c\eta X$ on X at level $\ell \leq X$, growing with X . This exponent cannot be absorbed into $X^{c\eta}$ for any absolute c .*

Proof. Per-character ([Theorem 1](#)): $N_\chi(1 - \eta, T) \ll \ell^{(\ell - 2)c_3\eta} T^{(\ell - 1)c_3\eta}$. Summing in A and ℓ :

$$\sum_{\ell \leq X} \sum_A N_{\chi_{\ell, A}}(1 - \eta, T) \ll T^{c_3\eta} \sum_{\ell \leq X} \ell^{c_3\eta\ell + O(1)} \leq X^{c_3\eta X + O(1)} T^{c_3\eta}.$$

The exponent $c_3\eta X$ is bounded by $c_3\eta$ only when $X = O(1)$, contradicting any $X \rightarrow \infty$ uniform statement. $\square$

Theorem 6 (Artin descent does not help). *The induced representation $\rho_{\ell,A} = \text{Ind}_{K_\ell}^{\mathbb{Q}} \chi_{\ell,A}$ of dimension $\ell - 1$ on $G_{\mathbb{Q}}$ has Artin conductor $\text{cond}(\rho_{\ell,A}) = \ell^{\ell-2} \cdot N(\mathfrak{f})$. By the Bushnell–Henniart local pair-conductor bound applied at the wild-ramified place ℓ ,*

$$\log \text{cond}(\rho_{\ell_1,A_1} \otimes \tilde{\rho}_{\ell_2,A_2}) \leq (\ell_2 - 1) \log \text{cond}(\rho_{\ell_1,A_1}) + (\ell_1 - 1) \log \text{cond}(\rho_{\ell_2,A_2}) \asymp \ell_1 \ell_2 \log(\ell_1 \ell_2).$$

The cross-conductor exponent is identical to Theorem 4. Descent therefore provides no structural improvement; the V16 Cross-Field Rankin–Selberg Conductor Explosion is intrinsic to the family, not an artifact of the base field K_ℓ varying.

Proof of Theorem 3. A direct combination of Theorem 4, Theorem 5, and Theorem 6: any majorant route through PTW, Weiss, Thorner, Thorner–Zaman, Heath–Brown faces the super-polynomial off-diagonal cost (Theorem 4) or the super-polynomial summed-per-character cost (Theorem 5); the Artin descent maneuver yields the same cross-conductor explosion (Theorem 6). $\square$

5 Counterexample refutation: no Landau–Siegel zero

Theorem 7 (Refutation of Vandiver-residual counterexample). *For every prime ℓ with $3 \leq \ell < V_{\text{Vandiver}} = 1.63 \times 10^8$ and every $A \in \mathbb{F}_\ell^2 \setminus 0$, the Hecke L -function $L(s, \chi_{\ell,A})$ has no Landau–Siegel real zero in the standard sense $1 - \beta < c / \log \mathfrak{q}(\chi_{\ell,A})$ for c small absolute, modulo the Stark–Heath–Brown effective zero-free-region constant being uniform $\asymp 1/(\ell - 1)$ in the base-field degree.*

Proof. STEP 1: $\chi_{\ell,A}$ IS NOT REAL. $\chi_{\ell,A}$ has exact order ℓ (by rational Kummer independence of $\{2, 3\}$ over $\mathbb{Q}(\zeta_\ell)$). A real character has order dividing 2; since $\gcd(\ell, 2) = 1$ for $\ell \geq 3$, $\chi_{\ell,A}^2 \neq 1$, so $\chi_{\ell,A} \neq \overline{\chi_{\ell,A}}$.

STEP 2: THE ARAMATA–BRAUER FACTORIZATION OF $\zeta_{K_{\ell,A}}$ HAS NO REAL KUMMER-FAMILY COMPONENT. The Galois closure $\tilde{K}_{\ell,A}/\mathbb{Q}$ is metacyclic of order $\ell(\ell - 1)$, with the cyclotomic Galois group $(\mathbb{Z}/\ell)^*$ acting non-trivially on $\text{Gal}(K_{\ell,A}/K_\ell) \simeq \mathbb{Z}/\ell$ via the Kummer pairing. Decomposing:

$$\zeta_{K_{\ell,A}}(s) = \zeta(s) \cdot \prod_{j=1}^{\ell-2} L(s, \eta_j) \cdot L(s, \text{Ind}_{K_\ell}^{\mathbb{Q}} \chi_{\ell,A})$$

where η_j runs over the non-trivial Dirichlet characters mod ℓ , and $\text{Ind}_{K_\ell}^{\mathbb{Q}} \chi_{\ell,A}$ is the irreducible $(\ell - 1)$ -dimensional Artin representation whose L -function factors as $\prod_{j=1}^{\ell-1} L(s, \chi_{\ell,A}^j)$ over K_ℓ . Each $\chi_{\ell,A}^j$ for $1 \leq j \leq \ell - 1$ is non-real (Step 1). The only real character in this factorization is the unique quadratic Dirichlet character mod ℓ , the Legendre symbol $\eta_2 = \left(\frac{\cdot}{\ell}\right)$, which is *not* in the Kummer family $\{\chi_{\ell,A}\}_{A \in \mathbb{F}_\ell^2 \setminus 0}$.

STEP 3: VANDIVER RESIDUAL IS IN THE WRONG FAMILY. Even if $\text{Cl}(K_\ell^+)[\ell] \neq 0$ for some $\ell \geq V_{\text{Vandiver}}$, the resulting unramified Hecke character of K_ℓ would have order ℓ (or order 2 from a 2-class-field-tower contribution), not the Kummer order ℓ of $\chi_{\ell,A}$ with a specific α_A -radical conductor. These two character spaces are disjoint by ramification type: the Kummer character $\chi_{\ell,A}$ is ramified at all primes dividing $\alpha_A = 2^{A_1} 3^{A_2}$, whereas an unramified Hecke character of K_ℓ is unramified everywhere.

STEP 4: STARK–HEATH–BROWN ZERO-FREE REGION. By Stark 1974 / Heath–Brown 1992 (PLMS Lemma 2.4), for any non-real Hecke character χ of a number field K with $\chi^2 \neq 1$, $L(s, \chi)$ has no real zero in the range $1 - c_0 / \log \mathfrak{q}(\chi, T) < \beta < 1$ for $c_0 > 0$ effective absolute, provided the constant c_0 has the stated uniformity in K . Applying with $\chi = \chi_{\ell,A}$, $K = K_\ell$ gives the displayed bound for ℓ in the indicated range. $\square$

Remark 2. *The uniformity of the Stark–Heath–Brown constant c_0 in the base-field degree n_K is the single ingredient [Theorem 7](#) relies on beyond Steps 1–3 (which are unconditional structural facts). Standard expositions (Iwaniec–Kowalski Ch. 5.7, Heath–Brown 1992 §2) record this with constant c_0/n_K , which gives a zero-free region of width $\gg 1/((\ell - 1) \log \mathfrak{q})$. This is wider than a Siegel zero $o(1/\log \mathfrak{q})$ for ℓ bounded, confirming Step 4.*

6 The sharp dichotomy

Combining [Theorem 2](#) and [Theorem 3](#):

Theorem 8 (Sharp dichotomy at $\ell \log \ell \asymp \log Q$). *Let $\eta = c/\log Q$ for absolute $c > 0$, $T \leq (\log Q)^{B'}$.*

- (a) *For $\ell \log \ell \leq \log Q$, the joint family bound $\sum_{\ell} \sum_A N_{\chi_{\ell,A}}(1 - \eta, T) \ll (\log Q)^C$ holds unconditionally.*
- (b) *For $\ell \leq (\log Q)^B$ with $B > 1$, the same bound does not follow from any combination of PTW 2020, Weiss 1980/1983, Thorner 2017, Thorner–Zaman 2019/2021, Heath–Brown 1992.*

The threshold $\ell \log \ell \asymp \log Q$ corresponds to $\ell \asymp \log Q / \log \log Q$.

Remark 3 (Open seam). *The seam $\log Q / \log \log Q < \ell < \log Q$ (equivalently $1 < \ell \log \ell / \log Q < \log \log Q \cdot \log \log \log Q^{-1}$) is neither covered by [Theorem 2](#)’s strict unlock nor by [Theorem 3](#)’s strict no-go. In this seam the trivial concatenation gives polylog factors growing slowly in Q , but not absorbable into Q^ε for arbitrary fixed ε . Closing this seam unconditionally would require either a sub-multiplicative refinement of Bushnell–Henniart for the cyclotomic-Kummer family, or a new joint-field large sieve via moduli-of-cyclotomic-fields methods.*

Remark 4 (Relation to EG203). *The Erdős–Graham Problem #203 (unsolved) reduces, under the V_4 ship-bundle hypothesis stack, to the high-seam PHNC instance. [Theorem 2](#) unlocks PHNC unconditionally for $\ell \leq \log Q / \log \log Q$, which combined with the remaining KRWS, KRC, RTHLS, Kummer-BV, BVK stack would yield a conditional resolution of EG203 in the strict polylog regime. [Theorem 3](#) forecloses an unconditional resolution at $\ell \asymp (\log Q)^B$ for $B > 1$ via the named analytic toolkit alone.*

7 Methodology and the next mile

The present results were produced by an Oracle-pipeline-structured swarm comprising V16 (7 Phase-C lemma slots L0–L5 + E3, 1 Phase-D R667 INTERRUPTER, 1 BORG synthesis) and V17 (4 Phase-C lemma slots WTZ-A/B + ARTIN + CTREX, 1 Phase-D R667 INTERRUPTER, 1 BORG synthesis).

- V16 produced the JANG no-go ([Theorem 3](#)), Cross-Field Conductor Explosion barrier ([Theorem 4](#)), and the conditional Bridge corollary.
- V17 verified the Thorner–Zaman base-field discriminant dependence as additive-in-the-exponent, upgrading the conditional Bridge to the unconditional sharp cutoff ([Theorem 2](#)). V17 also (a) refuted the Vandiver-residual counterexample heuristic via the Aramata–Brauer factorization and Kummer-character-order argument ([Theorem 7](#)), and (b) proved the Artin descent provides no structural rescue ([Theorem 6](#)).

Open seam research target. The seam $\log Q / \log \log Q < \ell < \log Q$ remains the only unresolved range for joint-family log-free ZDE in the cyclotomic Kummer setting via current methods. A complete resolution of this seam, combined with the remaining EG203 hypothesis stack, would yield an unconditional resolution of Erdős–Graham Problem #203.

Note on cite-verification. Several citations to Thorner–Zaman 2019 (or 2021) and PTW 2020 specific proposition numbers in this paper were synthesized from the proof structure rather than line-by-line verified against the published papers. These are marked in the methodology trace. The structural arguments (Steps 1–3 of [Theorem 7](#), Bushnell–Henniart bookkeeping in [Theorem 4](#), Halász–Montgomery sourcing in [Theorem 1](#)) are unconditional.

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